

PROBLEM SET 4—SOLUTIONS

ECON301—ECONOMETRICS I

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions: Due on 13/3/2020 before 12 PM in the box outside my office. You are encouraged to work in groups but each student is required to submit his or her own work. If you worked with a group, please indicated the names of each member in your submission. Don't forget to attach the cover sheet. For the computer exercises, write your answer in here and include a printout of the R code that you used to answer the exercise.

1. Problem 1, chapter 3.

Solution

- (a) *hsperc* is defined so that the smaller it is, the lower the student's standing in high school. Everything else equal, the worse the student's standing in high school, the lower his/her expected college GPA.
- (b) Just plug these values into the equation

$$\widehat{colgpa} = 1.392 - .0135(20) + .00148(1050) = 2.676.$$

- (c) The difference between A and B is simply 140 times the coefficient on *sat*, because *hsperc* is the same for both students. So A is predicted to have a score $.00148(140) \approx .207$ higher.
- (d) With *hsperc* fixed, $\widehat{colgpa} = .00148\Delta sat$. Now, we want to find Δsat such that $\Delta \widehat{colgpa} = .5$, so $.5 = .00148(\Delta sat)$ or $\Delta sat = .5/ (.00148) \approx 338$. Perhaps not surprisingly, a large ceteris paribus difference in SAT score—almost two and one-half standard deviations—is needed to obtain a predicted difference in college GPA of a half a point.

2. Problem 3, chapter 3.

Solution

- (a) If adults trade off sleep for work, more work implies less sleep (other things equal), so $\beta_1 < 0$.
- (b) signs of β_2 and β_3 are not obvious. One could argue that more educated people like to get more out of life, and so, other things equal, they sleep less ($\beta_2 < 0$). The relationship between sleeping and age is more complicated than this model suggests, and economists are not in the best position to judge such things.
- (c) Since *totwrk* is in minutes, we must convert five hours into minutes: $\Delta totwrk = 5(60) = 300$. Then sleep is predicted to fall by $.148(300) = 44.4$ minutes. For a week, 45 minutes less sleep is not an overwhelming change.
- (d) More education implies less predicted time sleeping, but the effect is quite small. If we assume the difference between college and high school is four years, the college graduate sleeps about 45 minutes less per week, other things equal.
- (e) Not surprisingly, the three explanatory variables explain only about 11.3% of the variation in sleep. One important factor in the error term is general health. Another is marital status and whether the person has children. Health (however we measure that), marital status, and number and ages of children would generally be correlated with *totwrk*. (For example, less healthy people would tend to work less.)

3. Problem 4, chapter 3.

Solution

- (a) A larger rank for a law school means that the school has less prestige; this lowers starting salaries. For example, a rank of 100 means there are 99 schools thought to be better.
- (b) $\beta_1 > 0, \beta_2 > 0$. Both LSAT and GPA are measures of the quality of the entering class. No matter

where better students attend law school, we expect them to earn more, on average. $\beta_3 > 0, \beta_4 > 0$. The number of volumes in the law library and the tuition cost are both measures of the school quality. (Cost is less obvious than library volumes, but should reflect quality of the faculty, physical plant, and so on.)

- (c) This is just the coefficient on GPA, multiplied by 100: 24.8%.
- (d) This is an elasticity; a one percent increase in library volumes implies a .095% increase in predicted median starting salary, other things equal.
- (e) It is definitely better to attend a law school with a lower rank. If law school A has a ranking 20 less than that of law school B, the predicted difference in starting salary is $100(.0033)(20) = 6.6\%$ higher for law school A.

4. Problem C1, chapter 3.

Solution

- (a) Probably $\beta_2 > 0$, as more income typically means better nutrition for the mother and better prenatal care.
- (b) On the one hand, an increase in income generally increases the consumption of a food, and *cigs* and *faminc* could be positively correlated. On the other, family incomes are also higher for families with more education, and more education and cigarette smoking tend to be negatively correlated. The sample correlation between *cigs* and *faminc* is about $-.173$, indicating a negative correlation.
- (c) The regressions without and with *faminc* are

$$\widehat{bwght} = 119.77 - .515cigs$$

$$n = 1388, R^2 = .023$$

and

$$\widehat{bwght} = 116.97 - .464cigs + .093faminc$$

$$n = 1,388, R^2 = .030.$$

The effect of cigarette smoking is slightly smaller when *faminc* is added to the regression, but the difference is not great. This is due to the fact that *cigs* and *faminc* are not very correlated, and the coefficient on *faminc* is practically small. (The variable *faminc* is measured in thousands, so \$10,000 more in 1988 income increases predicted birth weight by only .93 ounces.)

5. Problem C2, chapter 3.

Solution

- (a) The estimated equation is

$$\widehat{price} = -19.32 + .128sqft + 15.20bdrms$$

$$n = 88, R^2 = .632$$

- (b) Holding square footage constant, $\Delta \widehat{price} = 15.20 \Delta bdrms$, and so $\widehat{price}$ increases by 15.20, which means \$15,200.
- (c) Now $\Delta \widehat{price} = .128 + 15.20 \Delta bdrms = .128(140) + 15.20 = 33.12$, or \$33,120. Because the size of the house is increasing, this is a much larger effect than in (ii).
- (d) About 63.2%.
- (e) The predicted price is $-19.32 + .128(2,438) + 15.20(4) = 353.544$, or \$353,544.
- (f) From part (v), the estimated value of the home based only on square footage and number of bedrooms is \$353,544. The actual selling price was \$300,000, which suggests the buyer underpaid by some margin.

But, of course, there are many other features of a house (some that we cannot even measure) that affect price, and we have not controlled for these.